$C37_A1$

Changze Wu

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1 Q1

Proof: Assume, b is the base of the computer's floating point number system and t is the mantissa length. We only consider that the b is even

- case 1: when chopping: Consider the mantissa length of $(1+\epsilon)_b$ would be the # of significant digits of it. So when we consider the ϵ in chopping, it would be the number of $(.\underbrace{00...001}_{t-1})_b$, because there is a 1 before the dot. Thus, we conclude that the $\epsilon = (1 \times b^{1-t})$ as wanted.

- case 2: when rounding : Consider the mantissa length of $(1+\epsilon)_b$ would be the # of significant digits of it. So when we consider the ϵ in rouning , we should consider the number$(.\underbrace{00...000}_{t-1} d_i)_b$. if $d_i \geq \frac{1}{2}b$, it will influence the t-1 -th 0; otherwise, chopping. Thus, the minimum valur of d_i is $\frac{1}{2}b$. According to that, the $\epsilon = \frac{1}{2}(.\underbrace{00...001}_{t-1})_b$, which is the $\frac{1}{2}b^{1-t}$, as wanted.

Conclude above all, we can easily get the consequence we want to prove. QED

2 Q2

Let $x, y \in \mathcal{R}$, then $Fl(x) = x(1-\delta_x)$, $Fl(y) = y(1-\delta_y)$

Division $\frac{x}{y}$, computer gives

$$
\begin{aligned}
Fl(\frac{Fl(x)}{Fl(y)}) &= [\frac{x(1-\delta_x)}{y(1-\delta_y)}](1-\delta_{xy}) \\
&= \frac{x(1-\delta_x)(1-\delta_{xy})}{y(1-\delta_y)} \\
&= \frac{x}{y} \cdot \frac{1-\delta_x-\delta_{xy}+\delta_x \cdot \delta_{xy}}{1-\delta_y} \\
&= \frac{x}{y} \cdot \frac{(1-\delta_x-\delta_{xy}+\delta_x \cdot \delta_{xy})(1+\delta_y)}{1-\delta_y^2} \\
&\approx \frac{x}{y} \cdot (1-\delta_x-\delta_{xy})(1+\delta_y) & |\delta| < \epsilon, \delta^2 \ll |\delta| \ll 1 \\
&= \frac{x}{y} \cdot (1+\delta_y-\delta_x\delta_y-\delta_x-\delta_{xy}-\delta_{xy}\delta_y) \\
&\approx \frac{x}{y} \cdot (1+\delta_y-\delta_x-\delta_{xy}) \\
&\approx \frac{x}{y} \cdot (1-\delta_{total}) & \delta_{total} = -\delta_y+\delta_x+\delta_{xy}
\end{aligned}
$$

Since $|\delta_x|, |\delta_y| \leq \epsilon \quad and \quad |\delta_{xy}| \leq \epsilon$, the total error δ_{total} is bounded by:$|\delta_{\text{total}}| \leq |\delta_x| + |\delta_y| + |\delta_{xy}| \leq \epsilon + \epsilon + \epsilon = 3\epsilon$

3 Q3

According to the topic, x is a floating-point machine number in a computer that has unit roundoff ϵ, so we can get $fl(x) = x$ at first round.

When we do $fl(x^2)$ we compute the $fl(x \cdot x)$ which may cause unit round off when we do $x \cdot x$, according to the topic $|\delta| \leq \epsilon$, we can simply estimate this error in $1 + \delta$ or $1 - \delta$, here we choose the $1 + \delta$ to suit for the topic.

Here is the process,

1) $fl(x) = x$,

2) $fl(x^2) = x^2 \cdot (1 + \delta_1)$,

3)$fl(x^3) = fl(x^2 \cdot x) = x^2 \cdot x \cdot (1 + \delta_1)(1 + \delta_2) = x^3 \cdot (1 + \delta_1)(1 + \delta_2)$

4)....

k)$fl(x^k) = x^k \cdot (1 + \delta_1)...(1 + \delta_{k-1})$,

with $|\delta| \leq \epsilon$, we can get that δ_i, with i = 1, ..., k-1 are very small, we can simply make them $\delta_1 \approx \delta_2 \approx ... \approx \delta_{k-1} = \delta$

Then we can conclude that $fl(x^k) = x^k \cdot (1 + \delta)^{k-1}$, as wanted.

4 Q4

Considering the overflow, underflow and cancellation:

- Overflow and Underflow: When the coefficients are very large or small, then b^2 or ac may too big or small to cause the overflow or underflow. Apart from that, the overflow or underflow may also be caused by formula of quadratic, such as when a is small, and the dividing will cause the result too big. In this way, here is a method to avoid the problem. For example, we can rescale the coefficients by dividing all three by the largest cofficient. This rescaling doesn't change the root of the quadratic equation, but now the largest coefficient is 1, so the overflow can be avoid when computing b^2 or ac. This method can't eliminate the possibility of underflow, but it prevents unnecessary underflow, which might otherwise occur when all three coefficients are very small.

- Cancellation: When $b^2 - 4ac$ is approximately equal to b^2, there might be a subtraction cancellation between $-b$ and $\sqrt{b^2 - 4ac}$. Thus, we can use the alternative formula to calculate the roots:

$$x = \frac{2c}{-b \mp \sqrt{b^2 - 4ac}}$$

This formula has the opposite sign pattern compared to the standard one. In this way, the problem of subtractive cancellation between -b and $\sqrt{b^2 - 4ac}$ is mitigated, as the formula avoids direct subtraction of two nearly equal quantities. However, it should be noted that cancellation within the square root term itself is harder to avoid without using higher precision arithmetic.

5 Q5

5.1 a

- $\|x\| \geq 0$ since $|x_i| \geq 0$ for all i . $\|x\| = 0$ if and only if all $x_i = 0$. holds.

- $\|\alpha x\| = \max\{|\alpha x_1|, |\alpha x_2|, \ldots, |\alpha x_n|\} = |\alpha| \max\{|x_1|, |x_2|, \ldots, |x_n|\} = |\alpha|\|x\|$, holds

- $\|x+y\| = \max\{|x_1+y_1|, |x_2+y_2|, \ldots, |x_n+y_n|\} \leq \max\{|x_1|, |x_2|, \ldots, |x_n|\} + \max\{|y_1|, |y_2|, \ldots, |y_n|\} = \|x\| + \|y\|$, holds.

Thus, a is a vector form.

5.2 b

- $\|x\| \geq 0$ since $|x_i| \geq 0$. However, $\|x\| = 0$ does not imply that all components of x are zero (specifically, x_1 could be non-zero).not hold.

Thus, b is not vector form.

5.3 c

- $\|x\| = n$, which is always positive regardless of the values of x. Assume, n is arbitrary number greater than 0, than no matter what x is, $\|x\|$ can't be 0. not holds.

Thus, c is not vector form

5.4 d

- $\|\alpha x\| = \sum_{i=1}^{n} |\alpha x_i|^3 = |\alpha|^3 \sum_{i=1}^{n} |x_i|^3 = |\alpha|^3 \|x\|$. This does not equal $|\alpha|\|x\|$. not holds

Thus, d is not vector form

5.5 e

- $\|x+y\| = [\sum_{i=1}^{n} |x_i + y_i|^{\frac{1}{2}}]^2$

Assume x = (0, 1), y = (1, 0). we can get the result: $r_1 = ((1+0)^{\frac{1}{2}} + (0+1)^{\frac{1}{2}})^2 = 4$

When we do separately, $r_2 = \|x\| + \|y\| = (1^{\frac{1}{2}} + 0^{\frac{1}{2}})^2 + (0^{\frac{1}{2}} + 1^{\frac{1}{2}})^2 = 1 + 1 = 2$

We get that $r_1 > r_2$, which doesn't satisfy the $\triangle$ inequality. not holds

Thus, e is not vector form